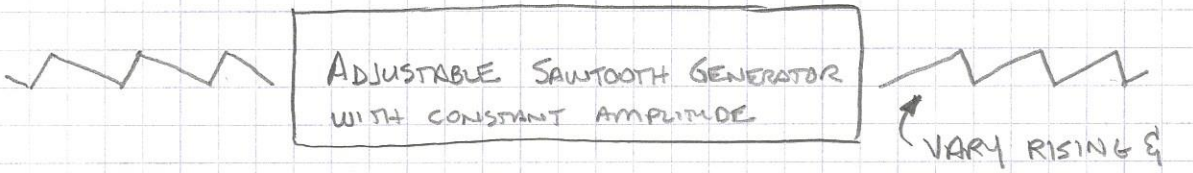
CIRCUIT FUN



USING :

- CURRENT SOURCES
- DIODE SWITCH BRIDGE
- COMPARATOR WITH HYSTERESIS

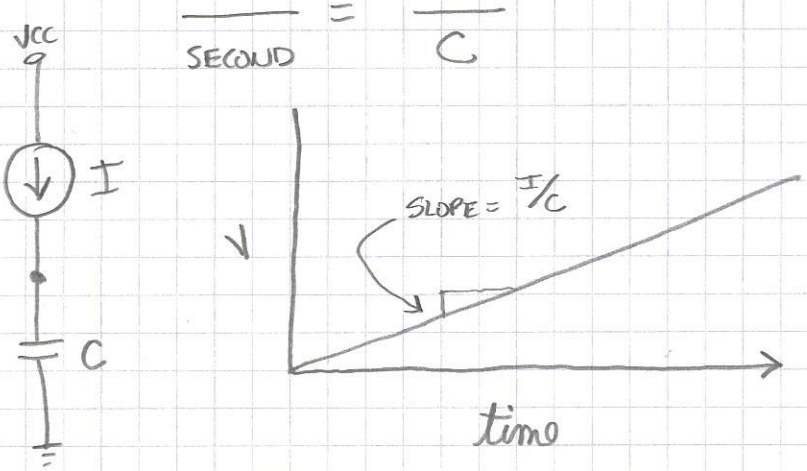
W2AEW

UNDERLYING PRINCIPLE:

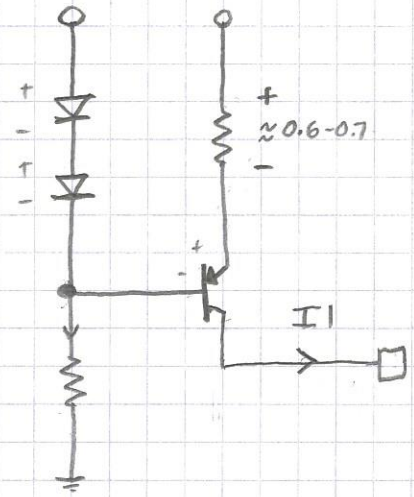
- LINEAR VOLTAGE RAMP OCCURS WHEN A CONSTANT CURRENT CHARGES OR DIS-CHARGES A CAPACITOR

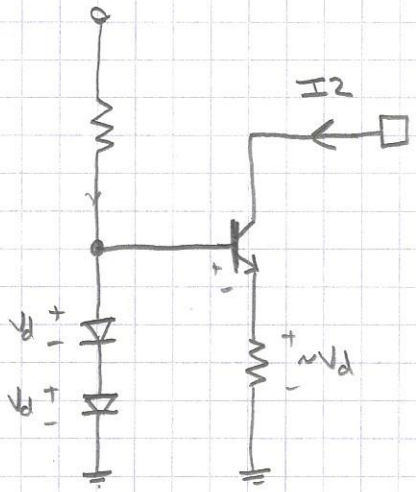
$$i = \frac{dV}{dt} * C$$

$$\frac{\text{VOLTS}}{\text{SECOND}} = \frac{I}{C}$$



CURRENT SOURCES:





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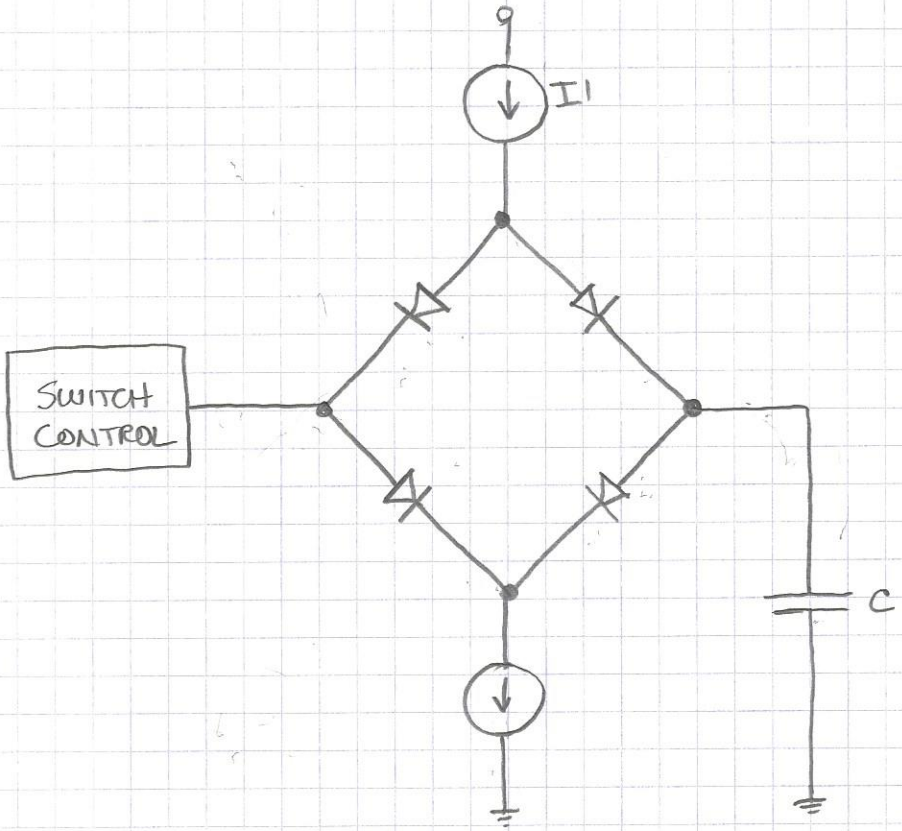
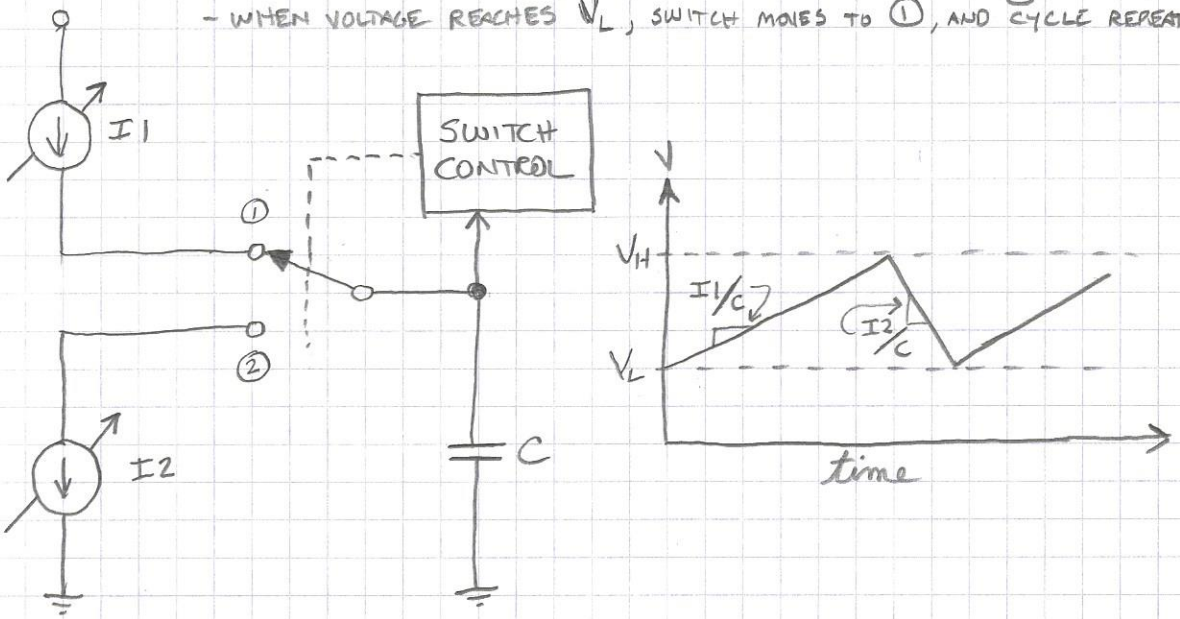
SPDT CURRENT SWITCH USING A DIODE BRIDGE



FUNCTIONAL CIRCUIT DIAGRAM

- SWITCH IN POSITION ①, VOLTAGE ON CAP RISES AT $\frac{I1}{C}$ VOLTS/SEC.
- WHEN VOLTAGE REACHES V_H, SWITCH CONTROLS MOVES SWITCH TO ②
- SWITCH IN POSITION ②, VOLTAGE ON CAP FALLS AT $\frac{I2}{C}$ VOLTS/SEC.
- WHEN VOLTAGE REACHES V_L, SWITCH MOVES TO ①, AND CYCLE REPEATS



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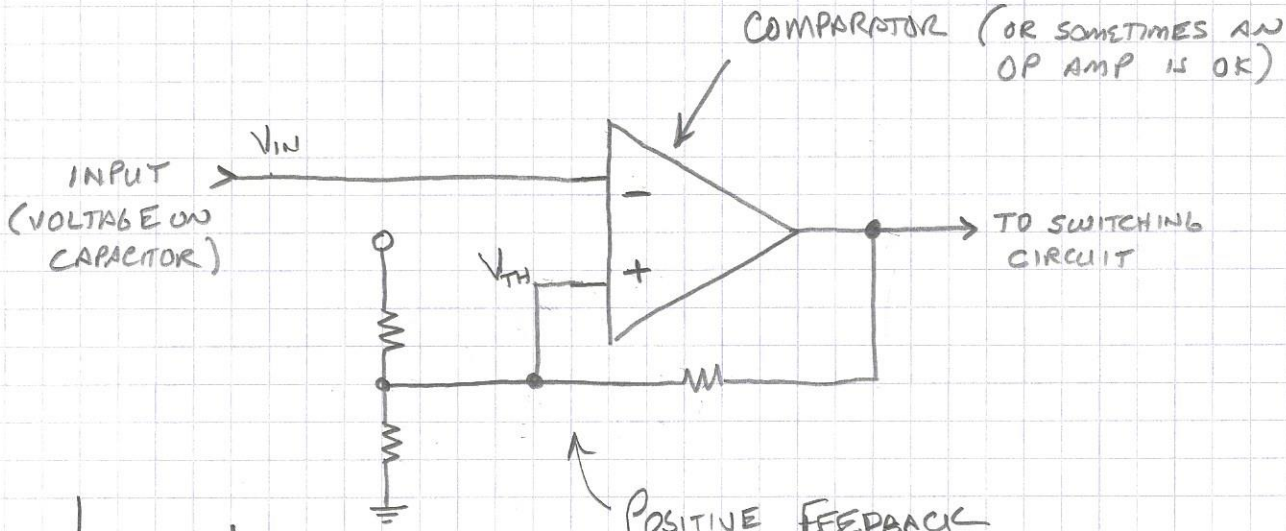
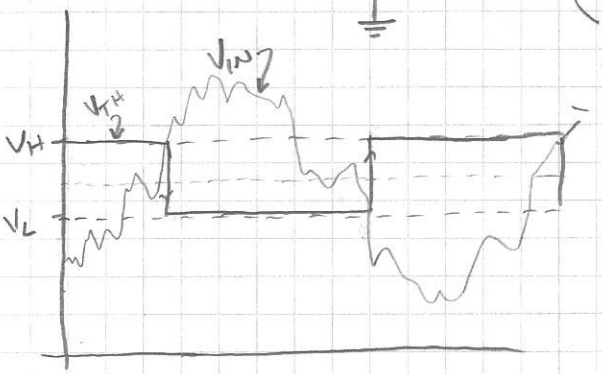
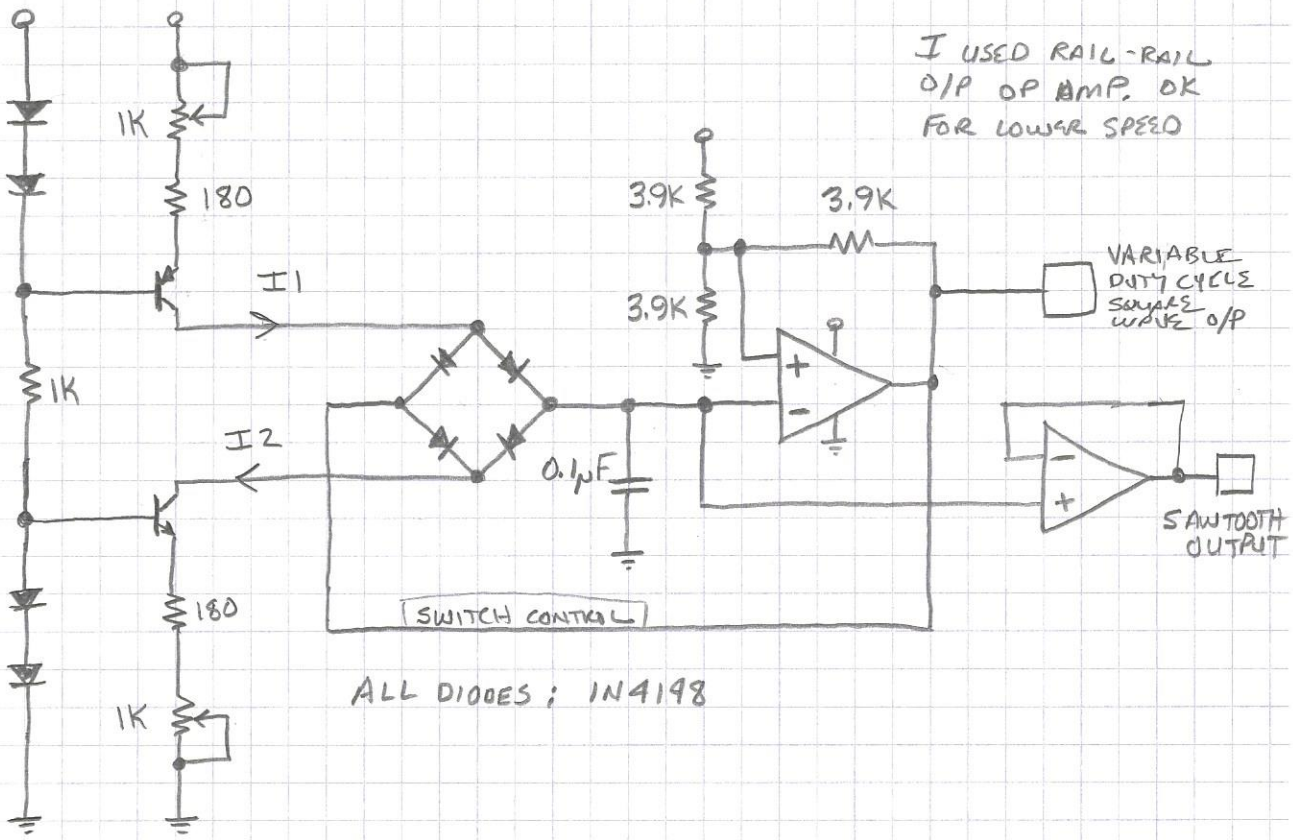
SWITCH CONTROL - COMPARATOR WITH HYSTERESIS





- CAUSES THRESHOLD TO CHANGE DEPENDING ON THE OUTPUT STATE

- THIS IS HOW SCHMITT TRIGGER CIRCUITS WORK (NOT ADJUSTABLE)

ADJUSTABLE SAWTOOTH GENERATOR